
The Memory Overhead: Rethinking Memory Scalability in Agent Systems

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Expanding an agent’s accessible memory does not make the task itself harder, but
2 it can reduce budgeted reliability under a fixed retrieval-call budget (agent-issued
3 external memory API calls). We study this setting by increasing irrelevant acces-
4 sible memory while keeping the task-relevant evidence fixed. We introduce an
5 evaluation protocol based on $\text{Pass}@B$, retrieval cost, failure decomposition, and
6 the scale at which budgeted reliability breaks down, and apply it to LongMemEval
7 and LoCoMo under matched budgets and the same number of returned items at
8 each retrieval step. We evaluate HippoRAG and LiCoMemory, showing that larger
9 memory lowers budgeted reliability even when the evidence is unchanged, but the
10 dominant failure mode depends on the agent model and memory interface: single-
11 pass retrieval mainly produces wrong answers within budget, whereas weaker
12 multi-hop settings more often run out of budget. Overall, budgeted reliability
13 at scale depends on whether the agent can recover the right evidence through the
14 memory interface under an agent-side interaction budget, not on memory storage
15 capacity alone.

16 1 Introduction

17 LLM agents have demonstrated strong capabilities in complex tasks by combining model reason-
18 ing with external tools, structured workflows, and autonomous environment interaction [Yao et al.,
19 2022, Wu et al., 2023, Zhou et al., 2023]. These agents operate over extended horizons, executing
20 multi-step plans, accumulating observations, and refining actions through iterative feedback [Yao
21 et al., 2022, Wang et al., 2023, Zhou et al., 2023]. To support long-term interaction with the environ-
22 ment, agents are equipped with a *memory system* that persists information across sessions and makes
23 relevant context available at inference time [Packer et al., 2023, Xu et al., 2025, Li et al., 2025].

24 Prior work on agent memory primarily falls into two categories. *Systems* research is concerned
25 with designing memory backends that enable efficient retrieval of relevant evidence [Fang et al.,
26 2025, Li et al., 2025, Jiménez Gutiérrez et al., 2024, Huang et al., 2025]. *Evaluation* research, on
27 the other hand, typically assesses accuracy or retrieval quality using a fixed memory snapshot [Wu
28 et al., 2024, Maharana et al., 2024, He et al., 2026]. However, in real-world deployments, agents’
29 memories expand over time as additional sessions and observations accumulate [Packer et al., 2023,
30 Wu et al., 2024, Maharana et al., 2024, Hu et al., 2025]. Despite this, previous research has not
31 asked how the continual growth of accessible memory impacts agent behavior. To address this, we
32 study agent behavior under **memory at scale**: specifically, by increasing the amount of irrelevant
33 accessible memory while keeping task-relevant evidence fixed. This design separates memory scale
34 from task difficulty [Wu et al., 2024], enabling us to study how scale affects agent behavior without
35 altering the task or its relevant evidence, as illustrated by the evidence-preserving scaling setup in
36 Figure 1(a).

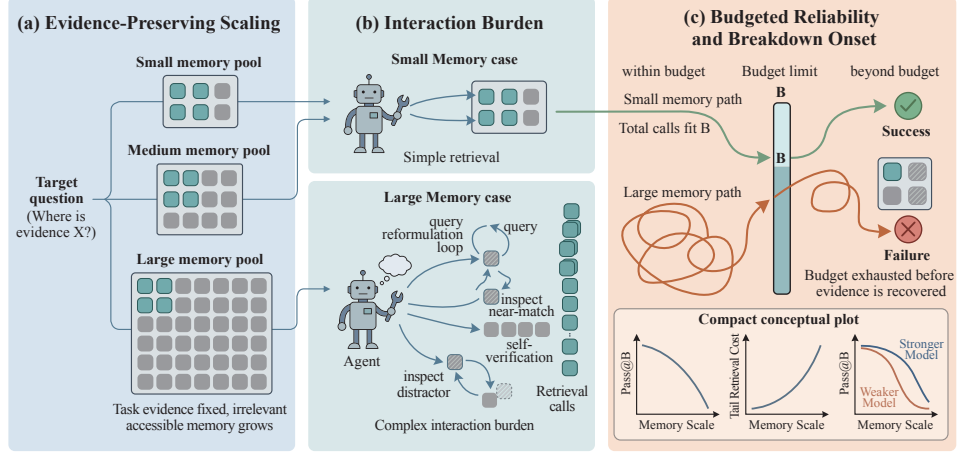


Figure 1: **Overview of memory at scale.** (a) *Evidence-preserving scaling*: task-relevant evidence is held fixed while irrelevant accessible memory grows. (b) *Interaction burden*: larger memories can trigger long-tail retrieve to verify loops (more agent-issued memory calls). (c) *Budgeted reliability and breakdown onset*: under a fixed agent-side retrieval-call budget, burden and evidence dilution translate into failures, shifting the usable-scale boundary across agent–memory systems.

37 In this setting, as accessible memory grows, the correct evidence becomes harder to identify among
 38 many irrelevant sessions. The form of this difficulty depends on the memory interface. Some agent–
 39 memory pairs remain effectively single-pass but become more likely to produce a wrong answer
 40 within budget, whereas others may need multiple rounds of retrieval and verification that increase
 41 the chance of budget exhaustion (Figure 1(b)). In both cases, the agent must decide whether to
 42 continue searching or to commit under uncertainty. Under finite interaction budgets, this pressure
 43 can reduce budgeted reliability even when the required evidence is present.

44 To systematically evaluate agent behavior under memory at scale, we introduce a budget-aware eval-
 45 uation protocol. We apply the protocol to LongMemEval [Wu et al., 2024] and LoCoMo [Maharana
 46 et al., 2024], and define a unified evaluation taxonomy spanning *flat*, *planar*, and *hierarchical* mem-
 47 ory interfaces [Hu et al., 2025]. The taxonomy specifies how each interface is evaluated under the
 48 same accessible history $\mathcal{H}^{(s)}$, retrieval-call budget B , shared scale levels, and the same number of
 49 returned items at each retrieval step. We report results for two contrasting hierarchical interfaces,
 50 HippoRAG and LiCoMemory [Jiménez Gutiérrez et al., 2024, Huang et al., 2025]: a largely single-
 51 pass retrieval interface and a multi-hop graph-retrieval interface. Flat and planar systems follow
 52 the same protocol and are included as controlled comparisons when reported. The protocol centers
 53 on budgeted reliability, failure source, interaction burden, and breakdown onset. We focus on
 54 agent-side online burden under a fixed retrieval-call budget, rather than memory-backend through-
 55 put, storage engineering, or offline indexing scalability. To quantify performance, we use $\text{Pass}@B$
 56 as the primary budgeted-reliability metric and define the breakdown onset s^* as the first evaluated
 57 scale where budgeted reliability falls below a threshold. Our results show that memory growth does
 58 not expose a single common failure mode. Instead, the effect of memory growth depends on the
 59 memory interface. In a fixed-size single-pass retrieval setting, more irrelevant memory increases
 60 competition for top- k slots, making relevant evidence easier to miss. In a multi-hop interface, the
 61 same increase can lead to additional traversal and verification steps, as well as difficult stopping
 62 decisions. Therefore, scalability depends on whether the model can control the memory interface
 63 under budget, not on storage capacity alone.

64 Our contributions are summarized as:

- 65 • We identify a memory-at-scale failure mode in which budgeted reliability degrades as irrel-
 66 evant accessible memory grows, even when task-relevant evidence remains fixed.
- 67 • We introduce a unified budget-aware evaluation protocol for flat, planar, and hierarchi-
 68 cal memory interfaces on LongMemEval and LoCoMo, measuring budgeted reliability,
 69 retrieval cost, failure source, and the scale at which budgeted reliability breaks down under
 70 fixed retrieval budgets.

- Our results distinguish precision-limited failures from interaction-limited failures and show that performance at scale depends on the agent model and memory interface, not on storage capacity alone.

2 Related Work

Memory architectures and cost. Memory-augmented agents rely on mechanisms for storing, organizing, and retrieving past information. Existing systems cover a broad design space and differ mainly in the structure exposed at retrieval time. For example, flat memories, such as LightMem and MemOS, store chunks, summaries, or dialogue turns and typically rely on dense or hybrid retrieval [Fang et al., 2025, Li et al., 2025]. Planar memories, such as A-MEM and LinearRAG, introduce explicit links within memory and support multi-step lookup and composition [Xu et al., 2025, Zhuang et al., 2025]. Hierarchical memories, such as HippoRAG and LiCoMemory, expose multiple levels of abstraction and support retrieval across different granularities [Jiménez Gutiérrez et al., 2024, Huang et al., 2025]. Systems such as EverMemOS further make memory construction and maintenance explicit parts of the overall pipeline [Hu et al., 2026]. However, using memory also incurs cost. Some systems pay more in preprocessing and maintenance, while others shift more of the burden to inference time through repeated retrieval and control [Fang et al., 2025, Jiménez Gutiérrez et al., 2024, Huang et al., 2025, Hu et al., 2026]. This issue becomes more important as accessible memory grows. Therefore, in this work, the focus is on this agent-side cost, especially the online burden imposed by different memory interfaces as accessible memory grows.

Benchmarks and scalability. Evaluating memory at scale requires benchmarks that vary accessible memory in a controlled way. Recent benchmarks have begun to examine long-horizon memory under growing histories. For example, LongMemEval increases irrelevant history while keeping the required evidence fixed, while LoCoMo provides longer and more natural multi-session conversations [Wu et al., 2024, Maharana et al., 2024]. However, most current evaluations focus on accuracy, retrieval quality, latency, or total cost at a fixed operating point. Such designs provide limited insight into how memory systems behave as accessible memory grows [He et al., 2026, Jiang et al., 2026]. In practice, scalability is a basic condition for usability. In this work, LongMemEval and LoCoMo are used as controlled testbeds to study how growing irrelevant accessible memory affects retrieval burden, budgeted reliability under fixed budgets, and the boundary of usable scale.

3 Methodology

This section formalizes the agent–memory evaluation protocol under controlled memory scaling. We first define agent interaction with an external memory under growing irrelevant content and fixed retrieval budgets (§3.1). Then, we introduce agent-centric, budget-aware metrics for assessing budgeted reliability, interaction cost, failure regimes, and breakdown onset as memory scales (§3.2).

3.1 Agent–Memory Interaction Under Scale

We study **memory scalability** as an *agent-side* property: how an agent model A interacts with a memory interface $\mathcal{M}$ as accessible memory grows under evidence-preserving scaling. For a query $q \sim \mathcal{D}$ and scale condition $s \in \mathcal{S}$, let $\mathcal{H}^{(s)}(q)$ denote the accessible history at scale s . We write $\mathcal{H}^{(s)}$ when the context is clear.

Evidence-preserving memory scaling. Let $y(q)$ denote the gold answer and let $E(q)$ denote the set of task-relevant evidence sessions for q . A family $\{\mathcal{H}^{(s)}(q)\}_{s \in \mathcal{S}}$ is *evidence-preserving* if for every s we have $E(q) \subseteq \mathcal{H}^{(s)}(q)$, the set $E(q)$ is unchanged across s , and $\mathcal{H}^{(s)}(q) \setminus E(q)$ contains only additional irrelevant sessions.

Agent–memory trajectories. Let $\mathcal{A}_{\text{ext}}(\mathcal{M})$ denote the set of agent-issued external memory API calls that count toward the retrieval-call budget. A rollout is

$$\tau = (o_1, a_1, o_2, a_2, \dots, o_T, a_T), \quad \tau \sim P(\tau \mid q, A, \mathcal{M}, \mathcal{H}^{(s)}(q)),$$

where o_t and a_t denote observations and actions, and t indexes trajectory steps. Define the retrieval-call count

$$R(\tau) := \sum_{t=1}^T \mathbf{1}[a_t \in \mathcal{A}_{\text{ext}}(\mathcal{M})],$$

and let $S(\tau) \in \{0, 1\}$ denote whether rollout τ returns a correct final answer.

Table 1: **Memory scales** used for LongMemEval and LoCoMo. Task-relevant evidence and query-type balance are fixed across scale because the same query set is evaluated at every s_k . LongMemEval uses the fixed2k construction by adding irrelevant sessions, while LoCoMo materializes the same $N_{\text{irr}}(s_k)$ values as distractor fragments. Approximate token counts refer to raw LongMemEval accessible history, not final prompt tokens.

Scale	N_{irr}	LongMemEval fixed2k	LoCoMo materialization	Raw LME tokens
s_0	0	evidence sessions only (1.225 avg.)	original dialogue	4.2K
s_1	100	+100 fixed2k irrelevant sessions	+100 distractor fragments	268K
s_2	200	+200 fixed2k irrelevant sessions	+200 distractor fragments	523K
s_3	300	+300 fixed2k irrelevant sessions	+300 distractor fragments	757K
s_4	400	+400 fixed2k irrelevant sessions	+400 distractor fragments	1.03M

In our experiments, $\mathcal{S} = \{s_0, \dots, s_4\}$ is a shared discrete ladder with matched irrelevant-session counts across benchmarks (Table 1). Budget IDs and the returned-item parity constraint are fixed within each system family, specified in § 4. Meanwhile, for graph-based systems, we use top- $k = 12$, and a returned item is a model-visible chunk-level evidence unit, not an internal graph node, path, or provenance session ID.

3.2 Agent-Centric Metrics for Memory Scalability

To characterize agent-memory behavior as s increases, we ask four questions: (1) Does budgeted reliability hold under a fixed interaction budget? (2) Does interaction burden grow with scale? (3) Which failure mode accounts for any budgeted-reliability drop? (4) At what scale does budgeted reliability break down?

3.2.1 Does budgeted reliability hold under a fixed interaction budget?

Definition 1 (Budgeted reliability — Pass@B). Let $R(\tau)$ be the retrieval-call count defined above. Define the per-rollout success indicator

$$\text{pass}_B(\tau) := \mathbf{1}[S(\tau) = 1 \wedge R(\tau) \leq B].$$

Budgeted reliability under system $(A, \mathcal{M})$ at scale s is

$$\text{Pass@B}(A, \mathcal{M}, s) := \mathbb{E}_{q \sim \mathcal{D}} \mathbb{E}_{\tau \sim P(\cdot | q, A, \mathcal{M}, \mathcal{H}^{(s)}(q))} [\text{pass}_B(\tau)]. \quad (1)$$

The $\text{pass}_B(\tau) \in \{0, 1\}$ is a per-rollout binary indicator and Pass@B is its expectation over the joint distribution of tasks and rollouts. In practice, we estimate Pass@B by averaging $\text{pass}_B(\tau)$ over evaluated tasks. A scalable system should maintain high Pass@B across $s_0 \rightarrow s_4$ under the same B .

3.2.2 Does interaction burden grow with scale?

Pass@B captures the outcome but not its cause: a drop could arise from heavier retrieval demand or from more within-budget errors, and both compress Pass@B equally. We therefore track how the retrieval-call distribution shifts with s .

Definition 2 (Interaction burden — MedR, P90R). Let $P_{A, \mathcal{M}, s}$ denote the marginal distribution

of $R(\tau)$ under $q \sim \mathcal{D}$ and $\tau \sim P(\cdot \mid q, A, \mathcal{M}, \mathcal{H}^{(s)}(q))$. Define the p -th quantile

$$Q_p(A, \mathcal{M}, s) := \inf\{r \in \mathbb{N}_0 : P_{A, \mathcal{M}, s}(R(\tau) \leq r) \geq p\}.$$

The median and 90th-percentile retrieval counts are:

$$\text{MedR}(A, \mathcal{M}, s) := Q_{0.5}(A, \mathcal{M}, s), \quad \text{P90R}(A, \mathcal{M}, s) := Q_{0.9}(A, \mathcal{M}, s). \quad (2)$$

MedR tracks the typical retrieval cost while P90R tracks the tail, capturing long retrieve-verify episodes from repeated query reformulation or verification under many irrelevant sessions. Tail pressure can grow visible before the median shifts, making both quantiles necessary.

3.2.3 Which failure mode accounts for any budgeted-reliability drop?

Growing interaction burden does not identify the failure type. The agent may exhaust the budget before finishing (*interaction-limited*) or complete within budget but return a wrong answer (*precision-limited*). We decompose failures to separate these regimes.

Definition 3 (Failure decomposition — $p_{\text{exh}}, p_{\text{wrong}}$). The events $\{\text{pass}_B(\tau) = 1\}$, $\{R(\tau) > B\}$, and $\{R(\tau) \leq B \wedge S(\tau) = 0\}$ partition the sample space, giving the identity

$$\text{Pass@B}(A, \mathcal{M}, s) + p_{\text{exh}}(A, \mathcal{M}, s; B) + p_{\text{wrong}}(A, \mathcal{M}, s; B) = 1, \quad (3)$$

where

$$\begin{aligned} p_{\text{exh}}(A, \mathcal{M}, s; B) &:= \Pr_{q, \tau}[R(\tau) > B], \\ p_{\text{wrong}}(A, \mathcal{M}, s; B) &:= \Pr_{q, \tau}[R(\tau) \leq B \wedge S(\tau) = 0], \end{aligned} \quad (4)$$

with (q, τ) distributed according to $q \sim \mathcal{D}$ and $\tau \sim P(\cdot \mid q, A, \mathcal{M}, \mathcal{H}^{(s)}(q))$.

The partition identity means any drop in Pass@B must be absorbed by a rise in $p_{\text{exh}}(\cdot; B)$, $p_{\text{wrong}}(\cdot; B)$, or both. Systems where $p_{\text{exh}}(\cdot; B)$ rises are *interaction-limited* (exceeding the retrieval-call budget), and those where $p_{\text{wrong}}(\cdot; B)$ rises are *precision-limited* (wrong within budget). Same Pass@B drop can reflect either regime, making this decomposition essential for interpretation.

3.2.4 At what scale does budgeted reliability break down?

The decomposition characterizes failure at each fixed s . To summarize where budgeted reliability crosses a usability threshold as s grows, we compress the full budgeted-reliability curve into a single boundary statistic.

Definition 4 (Breakdown onset — $s_\alpha^*(A, \mathcal{M}; B)$). Fix a budgeted-reliability threshold $\alpha \in (0, 1)$. The breakdown onset is

$$s_\alpha^*(A, \mathcal{M}; B) := \min\{s_k \in \{s_0, \dots, s_4\} : \text{Pass@B}(A, \mathcal{M}, s_k) < \alpha\}, \quad (5)$$

with $s_\alpha^*(A, \mathcal{M}; B) = +\infty$ if no such s_k exists within the ladder.

A smaller $s_\alpha^*(A, \mathcal{M}; B)$ indicates earlier loss of usable scale under the same budget. s^* should always be read jointly with the failure decomposition: two systems with the same s^* may reach the boundary through entirely different mechanisms. We set $\alpha = 0.7$ in all primary results.

4 Experimental Setup

This section details the evaluation setup, specifying benchmarks, memory scale conditions, model-backend configurations, and retrieval-call budgets. All quantities follow the definitions in § 3.2, including Pass@B (Eq. 1), interaction burden (Eq. 2), failure decomposition (Eq. 4), and breakdown onset (Eq. 5).

LLM Model and Memory System Configurations. We evaluate three agent-model capacity tiers (see Appendix Table 4 for Qwen3, with additional model families in Appendix A.3) in combination with three memory system architectures: **flat** (e.g., LightMem, MemOS), **planar** (A-Mem, Linear-RAG), and **hierarchical** (HippoRAG, LiCoMemory). For full system details, see Appendix Table 10. For all experiments, the accessible history $\mathcal{H}^{(s)}$, memory scale s , retrieval-call budget B , and the number of returned items at each retrieval step are held constant across systems.

Benchmarks and Memory Scale Definition. To isolate the effect of memory scaling on retrieval, we follow the setup in § 3: for each benchmark, s indexes a discrete memory-scale condition, and the associated irrelevant-session count is $N_{\text{irr}}(s)$, with all task-relevant evidence held fixed. Thus, changes in performance as s increases reflect agent–memory interaction rather than increased task difficulty. For both LongMemEval and LoCoMo, we use a shared set of memory scales $\{s_0, \dots, s_4\}$, corresponding to equivalent irrelevant-session counts $N_{\text{irr}}(s_k)$ at each level, specified in Table 1. Here, s_0 denotes an *evidence-only memory* scenario—i.e., no irrelevant sessions are present. Due to page limit, we report LoCoMo external-validity results in Appendix A.2.

We evaluate all systems under a retrieval-call budget B and the memory-scale levels s defined in § 3. For comparability across memory systems, we evaluate each task on the same accessible history $\mathcal{H}^{(s)}$ and keep fixed how many items are returned to the agent at each retrieval step.

For each benchmark, we construct an accessible history $\mathcal{H}^{(s)}$ that contains: (i) all task-relevant evidence required by the benchmark, and (ii) $N_{\text{irr}}(s)$ irrelevant sessions sampled with dataset-level deduplication.

Evaluation Size and Scoring. For the main Qwen LongMemEval sweeps, each $(A, \mathcal{M}, s)$ condition contains 2,000 queries with one rollout per query, yielding 10,000 rollouts per model–memory sweep across the five scale levels and 60,000 LongMemEval rollouts across the two hierarchical interfaces. The LoCoMo evaluation uses 282 queries per $(A, \mathcal{M}, s)$ condition, giving 1,410 rollouts per model–memory sweep and 8,460 LoCoMo rollouts across the Qwen hierarchical-interface matrix. Confidence intervals use 1,000 bootstrap resamples over task identifiers, preserving task-level dependence. Final-answer correctness is taken from the saved binary evaluator label; in post-processing, the evaluator label matches the recorded success flag on all included rows.

Retrieval-Call Budgets. The retrieval-call budget B constrains the number of allowed retrievals per query and is applied post-hoc through $\text{pass}_B(\tau)$ and $\text{Pass}@B$ (Eq. 1) as well as the failure rates (Eq. 4). Primary results use $B_0 = 2$, a strict budget that allows an initial memory access plus one follow-up retrieval or verification step while still exposing long-tail interaction failures. Budget IDs and larger-budget diagnostics are summarized in Appendix Table 8 and Table 9; Figure 9 further reports budget-sensitivity curves for $B \in \{1, \dots, 5\}$ under the same scale ladder and returned-item constraints.

Supplementary Budgeted-Reliability Diagnostics (Appendix). We focus on budgeted reliability under a retrieval-call budget (i.e., $\text{Pass}@B$ with the agent-issued external memory API calls). For diagnostics beyond this primary metric, we additionally report (i) *budget elasticity* curves (how budgeted reliability changes as B increases) and (ii) *stage-aware token accounting* (online query-time tokens vs. amortized preprocessing-plus-query tokens) in Appendix A.4.

5 Results

Overview. We evaluate memory scalability using the experimental protocol defined in § 4 and the metrics introduced in § 3.2. The results demonstrate that, given fixed task evidence and a limited retrieval-call budget, budgeted reliability is governed by the interplay among the agent model, memory interface design, and interaction budget. Specifically, our findings highlight three key points. First, evidence-only budgeted reliability is not sufficient to characterize scaled-memory performance (§ 5.1). Second, scale-induced failures arise from distinct mechanisms: *precision-limited degradation*, where the agent produces a wrong answer within budget, and *interaction-limited degradation*, where the rollout exceeds the retrieval-call budget (§ 5.3). Third, the dominant failure mode depends on both the memory interface and model capability (tail interaction burden in § 5.2, failure decomposition in § 5.3, budgeted-reliability boundaries in § 5.4).

Table 2: **Results under $B_0 = 2$ for the two memory system interfaces on LongMemEval.** HippoRAG starts strong at s_0 but degrades at scale, while LiCoMemory is capability-dependent: Qwen3-8B is already below the budgeted-reliability threshold at s_0 , whereas Qwen3-32B/Qwen3-235B remain above threshold through s_4 (see also Figure 2). $\text{Pass}@B_0$ (Eq. 1) is reported as mean $\pm$ 95% CI (bootstrapped over tasks). Breakdown onset $s_{0.7}^*(A, \mathcal{M}; B_0)$ (Eq. 5) is the first scale where $\text{Pass}@B_0$ drops below $\alpha = 0.7$.

Agent model A	Memory system $\mathcal{M}$	$\text{Pass}@B_0@s_0$ (%) $\uparrow$	$\text{Pass}@B_0@s_4$ (%) $\uparrow$	Cost@ s_4 (P90R) $\downarrow$	Breakdown onset $s_{0.7}^*(A, \mathcal{M}; B_0)$ $\uparrow$
Qwen3-8B	HippoRAG	78.2 ± 1.9	58.2 ± 2.3 ($\downarrow$ 20.0)	1	100
	LiCoMemory	53.5 ± 2.2	52.2 ± 2.3 ($\downarrow$ 1.3)	5	–
Qwen3-32B	HippoRAG	82.6 ± 1.7	63.5 ± 2.1 ($\downarrow$ 19.1)	1	100
	LiCoMemory	78.6 ± 1.9	78.5 ± 1.9 ($\downarrow$ 0.1)	4	>400
Qwen3-235B	HippoRAG	84.8 ± 1.6	68.8 ± 2.0 ($\downarrow$ 16.0)	1	300
	LiCoMemory	81.4 ± 1.7	80.1 ± 1.7 ($\downarrow$ 1.3)	2	>400

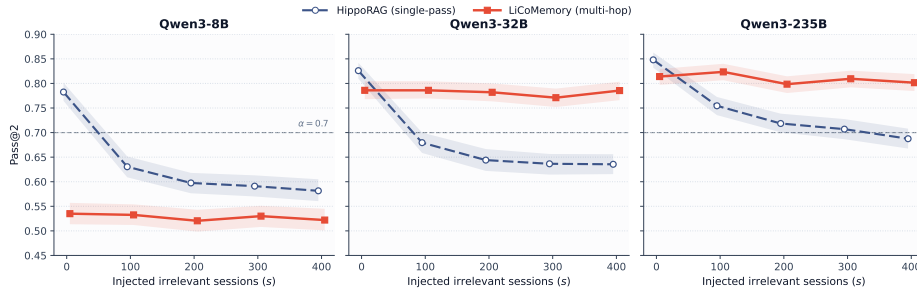


Figure 2: **Results under $B_0 = 2$ for the two memory system interfaces on LongMemEval.** HippoRAG degrades as memory scales; LiCoMemory exhibits two distinct regimes: with Qwen3-8B already below the budgeted-reliability threshold at s_0 but Qwen3-32B/Qwen3-235B remaining above threshold through the largest tested scale with breakdown onset $s_{0.7}^*(A, \mathcal{M}; B_0) > 400$ (Eq. 5). $\text{Pass}@B_0$ (Eq. 1) is reported as mean $\pm$ 95% CI (bootstrapped over tasks); the dashed line marks $\alpha = 0.7$.

5.1 Evidence-only budgeted reliability is not sufficient to characterize scaled-memory performance

We begin by asking whether evidence-only budgeted reliability at s_0 predicts performance once irrelevant sessions are introduced. Table 2 and Figure 2 show two pitfalls: evidence-only budgeted reliability can overestimate scaled budgeted reliability, and stable curves can hide already-broken regimes.

HippoRAG’s evidence-only result at s_0 overestimates scaled-memory budgeted reliability. After adding 400 irrelevant sessions, $\text{Pass}@B_0$ drops by 20.0, 19.1, and 16.0 points across the three Qwen tiers, pushing even the strongest model below the 70% threshold. Because the task-relevant evidence is unchanged across s , this drop reflects a scaling effect on evidence recovery rather than increased task difficulty.

LiCoMemory shows that flat curves can hide already-broken regimes. For Qwen3-8B, $\text{Pass}@B_0$ is already low at s_0 and stays below threshold as memory scales ($53.5\% \rightarrow 52.2\%$), so stability is not robustness. By contrast, Qwen3-32B and Qwen3-235B remain above threshold across all tested scales ($78.6\% \rightarrow 78.5\%$ and $81.4\% \rightarrow 80.1\%$). The contrast shows that scale robustness must be read jointly with the absolute budgeted-reliability level: a flat curve can indicate either genuine tolerance or failure that already occurs at baseline.

Overall, HippoRAG starts high but degrades sharply with scale, whereas LiCoMemory with Qwen3-8B looks stable only because it starts below threshold. We therefore next examine tail interaction burden, then failure decomposition, and finally breakdown onset to explain why these budgeted-reliability curves differ.

Table 3: **Failure decomposition for the two memory interfaces.** We report p_{exh} and p_{wrong} (Eq. 4) under the same budget $B_0 = 2$. Columns are grouped by failure event (interaction-limited: budget exhaustion; precision-limited: wrong within budget); red boxes highlight the larger failure component at s_4 .

Agent model A	Memory system $\mathcal{M}$	Interaction-limited (p_{exh} , %)		Precision-limited (p_{wrong} , %)	
		s_0	s_4	s_0	s_4
Qwen3-8B	HippoRAG	0.0	0.0	21.8	41.8
	LiCoMemory	37.2	38.6	9.3	9.2
Qwen3-32B	HippoRAG	0.0	0.0	17.4	36.5
	LiCoMemory	15.4	15.7	6.0	5.8
Qwen3-235B	HippoRAG	0.0	0.0	15.2	31.2
	LiCoMemory	1.9	2.0	16.8	17.9

5.2 Tail retrieval burden reveals model-dependent control cost

Evaluating budgeted reliability alone does not reveal whether failures come from poorer retrieval or heavier interaction. We therefore analyze the raw retrieval-call distribution $R(\tau)$ through MedR and P90R (Eq. 2); unlike $\text{Pass}@B_0$, these burden metrics are not truncated by the primary budget, so P90R(s_4) can exceed $B_0 = 2$.

The memory interface induces distinct agent-side interaction burden profiles. HippoRAG has a fixed one-call tail ($\text{P90R}(s_4) = 1$) across tiers, while LiCoMemory requires repeated memory access at scale (tail $\text{P90R}(s_4)$: $5 \rightarrow 4 \rightarrow 2$ as model capability increases). Thus, the main separation is not in the median case, but in how often difficult queries trigger extended interaction at the tail.

The memory interface determines interaction burden, and model capability determines whether it is manageable. LiCoMemory’s longer tail indicates that many queries require multiple rounds of memory access and verification before answering; stronger models reduce these extra calls, while weaker models more often accumulate budget pressure. We next ask whether this interaction burden manifests as budget-exceeding rollouts or wrong answers within budget.

5.3 Failure decomposition separates precision-limited and interaction-limited regimes

HippoRAG usually answers within budget, but increasingly answers incorrectly as memory grows. In Table 3, p_{exh} stays near zero across all Qwen tiers, while p_{wrong} rises from 21.8% to 41.8% (Qwen3-8B), 17.4% to 36.5% (Qwen3-32B), and 15.2% to 31.2% (Qwen3-235B) from s_0 to s_4 . This indicates that HippoRAG’s degradation is primarily a within-budget precision problem rather than an inability to stay within the call budget.

LiCoMemory with smaller models often needs more than the budgeted number of memory accesses. For Qwen3-8B, budget exhaustion dominates and remains high (p_{exh} : $37.2\% \rightarrow 38.6\%$) while p_{wrong} stays around 9%. For Qwen3-32B, exhaustion is lower but still persistent ($15.4\% \rightarrow 15.7\%$). For Qwen3-235B, exhaustion is largely mitigated ($1.9\% \rightarrow 2.0\%$), so the remaining error mass is mostly wrong answers within budget. The shift across model size shows that the same interface can move from interaction-limited to largely manageable once the agent can control repeated memory access more effectively.

5.4 Breakdown onset turns budgeted-reliability curves into usable-scale boundaries

Failure decomposition identifies *why* budgeted reliability drops; breakdown onset $s_{0.7}^*(A, \mathcal{M}; B_0)$ summarizes *where* usable budgeted reliability is lost. Under $B_0 = 2$ and $\alpha = 0.7$, it marks the first evaluated scale where $\text{Pass}@B_0$ falls below threshold. This converts the full budgeted-reliability curve into a practical memory-scale boundary that can be compared across systems under the same budget.

Usable scale depends on both the memory interface and model capability. LiCoMemory ranges from immediate breakdown with Qwen3-8B to $s_{0.7}^* > 400$ with Qwen3-32B and Qwen3-235B. HippoRAG improves more gradually, moving from breakdown at 100 irrelevant sessions in smaller

272 Qwen tiers to 300 for Qwen3-235B. Together with Table 3, these boundaries show that HippoRAG
273 is typically limited by rising within-budget errors, whereas LiCoMemory with Qwen3-8B is limited
274 by budget-exceeding rollouts. Additionally, breakdown onset operationalizes the trade-off between
275 accessible memory and budgeted reliability, turning diagnostic observations into actionable bound-
276 aries for practical system design.

277 **6 Conclusion**

278 We introduced a framework for evaluating agent–memory scalability under controlled, evidence-
279 preserving memory growth. By imposing a fixed retrieval-call budget and analyzing budgeted re-
280 liability, interaction burden, failure decomposition, and breakdown onset jointly, the framework
281 exposes how budgeted reliability changes as accessible memory scales. Across our experiments, the
282 dominant failure mode depends on both the memory interface and the agent model. More broadly,
283 these results suggest that scalable memory should be treated as an agent–memory systems prob-
284 lem, and they motivate future work on stronger agent-side memory control, more robust memory
285 interfaces, and evidence-level diagnostics.

References

- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022. URL <https://arxiv.org/abs/2210.03629>.
- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W White, Doug Burger, and Chi Wang. Autogen: Enabling next-gen llm applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023. URL <https://arxiv.org/abs/2308.08155>.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic web environment for building autonomous agents. *arXiv preprint arXiv:2307.13854*, 2023. URL <https://arxiv.org/abs/2307.13854>.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*, 2023. URL <https://arxiv.org/abs/2305.16291>.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023. URL <https://arxiv.org/abs/2310.08560>.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-mem: Agentic memory for llm agents. *arXiv preprint arXiv:2502.12110*, 2025. URL <https://arxiv.org/abs/2502.12110>.
- Zhiyu Li, Shichao Song, Hanyu Wang, Simin Niu, Ding Chen, Jiawei Yang, Chenyang Xi, Huayi Lai, Jihao Zhao, Yezhaohui Wang, Junpeng Ren, Zehao Lin, Jiahao Huo, Tianyi Chen, Kai Chen, Kehang Li, Zhiqiang Yin, Qingchen Yu, Bo Tang, Hongkang Yang, Zhi-Qin John Xu, and Feiyu Xiong. Memos: An operating system for memory-augmented generation (mag) in large language models. *arXiv preprint arXiv:2505.22101*, 2025. URL <https://arxiv.org/abs/2505.22101>.
- Jizhan Fang, Xinle Deng, Haoming Xu, Ziyan Jiang, Yuqi Tang, Ziwen Xu, Shumin Deng, Yunzhi Yao, Mengru Wang, Shuofei Qiao, Huajun Chen, and Ningyu Zhang. Lightmem: Lightweight and efficient memory-augmented generation. *arXiv preprint arXiv:2510.18866*, 2025. URL <https://arxiv.org/abs/2510.18866>.
- Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. Hipporag: Neurobiologically inspired long-term memory for large language models. *arXiv preprint arXiv:2405.14831*, 2024. URL <https://arxiv.org/abs/2405.14831>.
- Zhengjun Huang, Zhoujin Tian, Qintian Guo, Fangyuan Zhang, Yingli Zhou, Di Jiang, Zeying Xie, and Xiaofang Zhou. Licomemory: Lightweight and cognitive agentic memory for efficient long-term reasoning. *arXiv preprint arXiv:2511.01448*, 2025. URL <https://arxiv.org/abs/2511.01448>.
- Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. Longmemeval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*, 2024. URL <https://arxiv.org/abs/2410.10813>.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of llm agents. *arXiv preprint arXiv:2402.17753*, 2024. URL <https://arxiv.org/abs/2402.17753>.
- Zexue He, Yu Wang, Churan Zhi, Yuanzhe Hu, Tzu-Ping Chen, Lang Yin, Ze Chen, Tong Arthur Wu, Siru Ouyang, Zihan Wang, Jiaxin Pei, Julian McAuley, Yejin Choi, and Alex Pentland. Memoryarena: Benchmarking agent memory in independent multi-session agentic tasks. *arXiv preprint arXiv:2602.16313*, 2026. URL <https://arxiv.org/abs/2602.16313>.

333 Yuyang Hu, Shichun Liu, Yanwei Yue, Guibin Zhang, Boyang Liu, Fangyi Zhu, Jiahang Lin,
334 Honglin Guo, Shihan Dou, Zhiheng Xi, Senjie Jin, Jiejun Tan, Yanbin Yin, Jiongnan Liu, Zeyu
335 Zhang, Zhongxiang Sun, Yutao Zhu, Hao Sun, Boci Peng, Zhenrong Cheng, Xuanbo Fan, Jiaxin
336 Guo, Xinlei Yu, Zhenhong Zhou, Zewen Hu, Jiahao Huo, Junhao Wang, Yuwei Niu, Yu Wang,
337 Zhenfei Yin, Xiaobin Hu, Yue Liao, Qiankun Li, Kun Wang, Wangchunshu Zhou, Yixin Liu,
338 Dawei Cheng, Qi Zhang, Tao Gui, Shirui Pan, Yan Zhang, Philip Torr, Zhicheng Dou, Ji-Rong
339 Wen, Xuanjing Huang, Yu-Gang Jiang, and Shuicheng Yan. Memory in the age of ai agents, 2025.
340 URL <https://arxiv.org/abs/2512.13564>.

341 Luyao Zhuang, Shengyuan Chen, Yilin Xiao, Huachi Zhou, Yujing Zhang, Hao Chen, Qinggang
342 Zhang, and Xiao Huang. Linearrag: Linear graph retrieval augmented generation on large-
343 scale corpora. *arXiv preprint arXiv:2510.10114*, 2025. URL <https://arxiv.org/abs/2510.10114>.

345 Chuanrui Hu, Xingze Gao, Zuyi Zhou, Dannong Xu, Yi Bai, Xintong Li, Hui Zhang, Tong Li,
346 Chong Zhang, Lidong Bing, and Yafeng Deng. Evermemos: A self-organizing memory operating
347 system for structured long-horizon reasoning. *arXiv preprint arXiv:2601.02163*, 2026. URL
348 <https://arxiv.org/abs/2601.02163>.

349 Dongming Jiang, Yi Li, Songtao Wei, Jinxin Yang, Ayushi Kishore, Alys Zhao, Dingyi Kang,
350 Xu Hu, Feng Chen, Qiannan Li, and Bingzhe Li. Anatomy of agentic memory: Taxonomy and
351 empirical analysis of evaluation and system limitations. *arXiv preprint arXiv:2602.19320*, 2026.
352 URL <https://arxiv.org/abs/2602.19320>.

Table 5: **External validity on LoCoMo.** Endpoint $\text{Pass}@B_0$ (Eq. 1) is reported as mean $\pm$ 95% CI (bootstrapped over tasks) at the shared-scale endpoints s_0 and s_4 under $B_0 = 2$. The scaled endpoint uses the same irrelevant-session count convention as Table 1, so results are comparable under matched irrelevant-session loads.

Agent model A	Memory system $\mathcal{M}$	$\text{Pass}@B_0@s_0$ (%) $\uparrow$	$\text{Pass}@B_0@s_4$ (%) $\uparrow$	Drop (pp) $\downarrow$
Qwen3-8B	HippoRAG	58.2 ± 5.7	56.0 ± 6.0	2.1
	LiCoMemory	7.1 ± 3.2	5.7 ± 2.5	1.4
Qwen3-32B	HippoRAG	61.3 ± 6.0	58.5 ± 6.0	2.8
	LiCoMemory	58.5 ± 5.7	57.4 ± 6.0	1.1
Qwen3-235B	HippoRAG	64.2 ± 5.7	63.5 ± 5.7	0.7
	LiCoMemory	69.5 ± 5.3	66.3 ± 5.7	3.2

A Appendix

A.1 Qwen Model Diagnostics: Budgeted Reliability, Failure Modes, and Scalability

Table 4: **Qwen3 models used in our experiments.** We report the model name and native context length.

Model	Context
Qwen3-8B	32,768
Qwen3-32B	32,768
Qwen3-235B-A22B-Instruct-2507	262,144

Figures 3–4 provide supplementary visualizations of the endpoint results, retrieval-tail summaries, and failure decomposition, together with the gap between raw success and $\text{Pass}@B_0$ at scaled memory for LiCoMemory. Flat and planar systems follow the same evaluation protocol, but they are not used for the mechanism analysis unless their results are explicitly reported under the same (B_0, s) setting.

Budgeted Reliability and Tail Cost Across Scale and Model Size. Figure 3 adds two visual clarifications beyond the tables: the left panel distinguishes immediate low budgeted-reliability failure from stable high budgeted-reliability operation, and the right panel makes LiCoMemory’s tail compression with model size more visually explicit.

Failure Mode Decomposition and Budget-Induced Loss. Figure 4 adds the visual mechanism split behind the tables: HippoRAG remains dominated by p_{wrong} , whereas LiCoMemory is dominated by p_{exh} for smaller models and shifts toward p_{wrong} only at the strongest backbone. The right panel compares budget-induced loss across benchmarks, which is larger on LoCoMo for Qwen 8B and Qwen 32B and narrows with model size.

A.2 External Validity on LoCoMo

LongMemEval establishes a controlled framework by systematically introducing irrelevant sessions while maintaining fixed evidence. We include LoCoMo as an external-validity check, testing whether the diagnostic pattern persists in longer multi-session dialogues.

Results in Table 5 demonstrate that LoCoMo does not simply replicate the magnitude of performance declines observed in LongMemEval. Instead, it tests whether the previously identified evaluation outcomes hold in a qualitatively different memory context. For HippoRAG, performance drops on LoCoMo are consistently smaller across all Qwen model tiers than those observed on LongMemEval. For LiCoMemory, the effect of irrelevant sessions remains tied to model capacity: Qwen3-8B shows consistently low performance across both conditions ($7.1\% \rightarrow 5.7\%$), whereas higher-capacity models such as Qwen3-32B and Qwen3-235B maintain substantially stronger performance while still experiencing modest declines under additional irrelevant sessions ($58.5\% \rightarrow 57.4\%$ and $69.5\% \rightarrow 66.3\%$).

382 The implication for external validity is not that different benchmarks yield identical scale-induced
383 performance drops. Rather, the LoCoMo findings reinforce the general conclusion at the evaluation-
384 protocol level: additional irrelevant accessible memory can negatively affect budgeted reliability
385 even when task-relevant evidence remains unchanged, with specific outcomes depending on the
386 combination of model and memory interface.

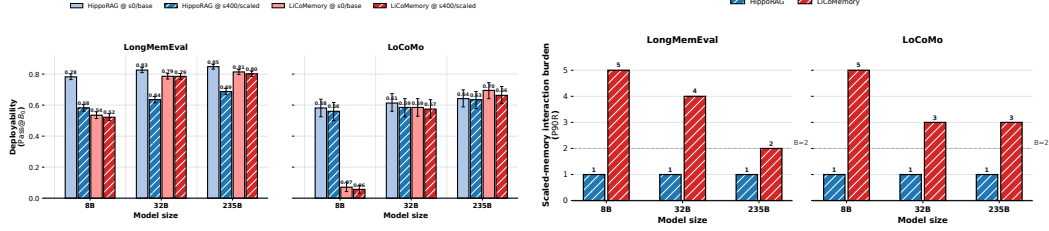


Figure 3: **Qwen endpoint budgeted reliability and scaled-memory tail burden.** Left: endpoint $\text{Pass}@B_0$ (Eq. 1) at s_0 and s_4 under $B_0 = 2$. Right: scaled-memory tail $\text{P90R}(s_4)$ (Eq. 2).

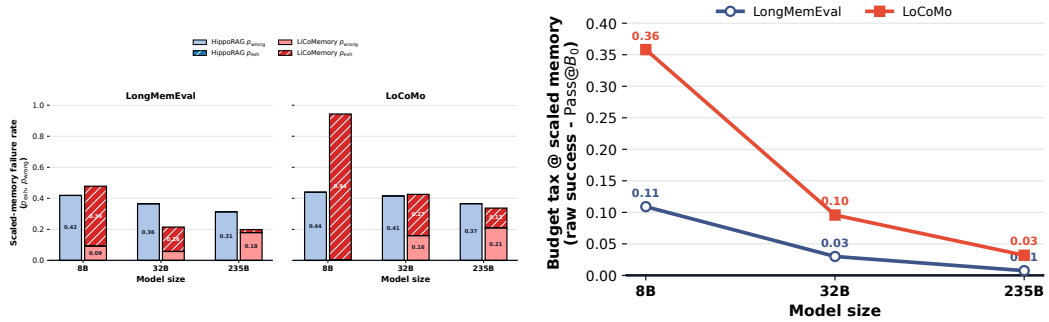


Figure 4: **Qwen failure decomposition and budget-induced loss at scaled memory.** Left: scaled-memory failure mass at s_4 , decomposed into p_{exh} and p_{wrong} (Eq. 4) under $B_0 = 2$. Right: LiCoMemory’s budget-induced loss at s_4 , defined as unbudgeted success $\Pr[S(\tau) = 1]$ minus $\text{Pass}@B_0$. HippoRAG is omitted from the right panel because its single-pass interface has negligible budget-exceeding mass throughout.

387 A.3 Supplementary Results: Additional Model Families

388 Table 6 summarizes the LLaMA-3.1 model variants included in our evaluation. This section extends
389 the main analysis with additional model-family diagnostics, first for LLaMA-3.1 and then for GPT-
390 OSS under LiCoMemory.

391 Figure 5 reports baseline (s_0) results for all three LLaMA-3.1 models (8B, 70B, 405B). Performance
392 improves consistently with model scale, as reflected in both $\text{Pass}@B_0$ and retrieval efficiency. The
393 hatched bars for 405B indicate observed values, highlighting the effect of increased backbone size
394 at the smallest irrelevant-session setting.

395 For a direct quantitative comparison, Table 7 presents endpoint results for the 8B and 70B models
396 at both baseline (s_0) and the largest fully observed shared scale (s_3). These results can be directly
397 compared to the Qwen series, illustrating both architecture- and scale-dependent trends in memory
398 retrieval performance.

Table 6: **LLaMA-3.1 models evaluated in this study.** We report the model name and native context length.

Model	Architecture	Native context
LLaMA-3.1-8B-Instruct	Dense	128,000
LLaMA-3.1-70B-Instruct	Dense	128,000
LLaMA-3.1-405B-Instruct	Dense	128,000

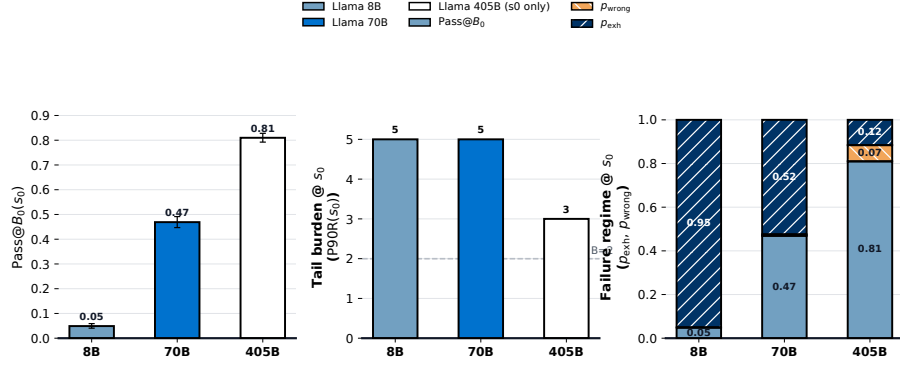


Figure 5: **LLaMA-3.1 s_0 baseline performance.** $Pass@B_0$, retrieval burden, and failure-regime summaries for 8B, 70B, and 405B at the smallest irrelevant-session setting. Hatched bars denote observed s_0 performance for 405B.

Table 7: **LLaMA-3.1 endpoints on LongMemEval.** Direct comparison on the fully observed shared endpoints s_0 (baseline) and s_3 for model sizes 8B and 70B. Green highlights the stronger 70B rows; red marks the 8B rows with severe exhaustion at s_0 .

Model	$Pass@B_0@s_0$	$Pass@B_0@s_3$	Cost@ s_0 (med/p90)	Exhaustion@ s_0
Llama 8B	0.049	0.047	5 / 5	0.951
Llama 70B	0.469	0.469	4 / 5	0.525

399 We also report GPT-OSS LiCoMemory runs on LongMemEval as an additional model-family diag-
 400 nostic. These runs use the same $Pass@B$ and retrieval-tail definitions as the main text and illustrate
 401 how scale, failure composition, and question-family structure vary for GPT-OSS backbones.

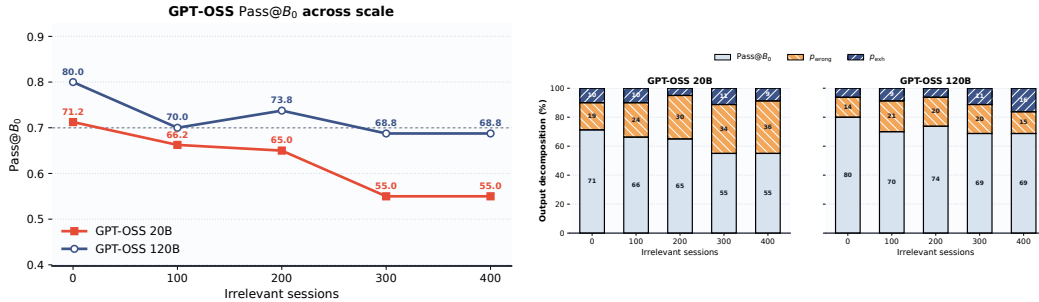


Figure 6: **GPT-OSS LiCoMemory scale diagnostics on LongMemEval.** Left: $Pass@B_0$ across the shared memory-scale ladder. Right: output decomposition across scale into $Pass@B_0$, within-budget wrong answers, and budget-exhaustion mass.

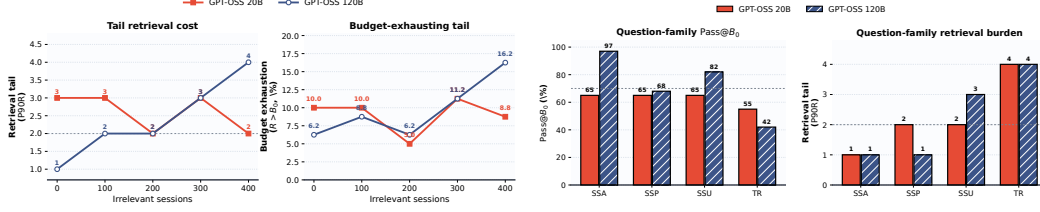


Figure 7: **GPT-OSS LiCoMemory retrieval and question-family diagnostics.** Left: retrieval-tail behavior across scale, including P90R and the fraction of budget-exhausting rollouts. Right: question-family Pass@ B_0 and retrieval-tail summaries.

A.4 Additional Diagnostics

These diagnostics provide mechanism-level support for the main claims and make the post-processing logic explicit.

Budget IDs and multi-budget appendix diagnostics. We report $B_0 = 2$ as the primary budget. Larger budgets are used only as supplementary diagnostics to quantify how much budgeted reliability can be recovered when the retrieval budget is relaxed under the same memory-scale ladder and returned-item constraints. Table 8 defines the named budget IDs used throughout the paper, and Table 9 gives the corresponding LongMemEval scaled-endpoint and breakdown summaries for the two hierarchical interfaces.

Table 8: **Retrieval-call budget IDs used in this paper.** B_0 is the primary budget in the main results; larger budgets are used for appendix diagnostics.

Budget ID	Value
B_0 (main)	2
B_1 (elastic)	3
B_2 (stress)	5

Table 9: **LongMemEval appendix budget summary for the two hierarchical interfaces.** Entries report scaled-endpoint budgeted reliability $\text{Pass@B}(A, \mathcal{M}, s_4)$ and the corresponding breakdown onset $s_{0.7}^*(A, \mathcal{M}; B)$ under the three named budgets, ordered as $B_0/B_1/B_2$. Entries are computed under the same evaluation pipeline and complement the left panel of Figure 9. Green highlights settings that remain above threshold through the observed scale ladder, while red marks the Qwen3-8B + LiCoMemory regime that stays below threshold even as budget increases.

Agent model A	Memory system $\mathcal{M}$	$\text{Pass@B}(s_4) (\%) \uparrow$	$s_{0.7}^*(A, \mathcal{M}; B) \uparrow$
Qwen3-8B	HippoRAG	58.1 / 58.1 / 58.1	$s_1 / s_1 / s_1$
	LiCoMemory	52.2 / 53.6 / 63.1	$s_0 / s_0 / s_0$
Qwen3-32B	HippoRAG	63.5 / 63.5 / 63.5	$s_1 / s_1 / s_1$
	LiCoMemory	78.5 / 79.0 / 81.5	$> s_4 / > s_4 / > s_4$
Qwen3-235B	HippoRAG	68.8 / 68.8 / 68.8	$s_4 / s_4 / s_4$
	LiCoMemory	80.2 / 80.9 / 80.9	$> s_4 / > s_4 / > s_4$

Supplementary token-cost accounting. The core protocol uses a retrieval-call budget rather than a token budget. We therefore report token expenditure only as a supplementary systems diagnostic in this appendix. In particular, the right panel of Figure 9 separates online query-time tokens (q_1) from amortized preprocessing-plus-query tokens ($q_0 + q_1$), so that cost placement in the pipeline can be inspected without conflating token accounting with the budgeted-reliability metric Pass@B .

Qualitative evidence source. Figure 8 provides a representative matched LongMemEval pair under LiCoMemory at $s = 400$.

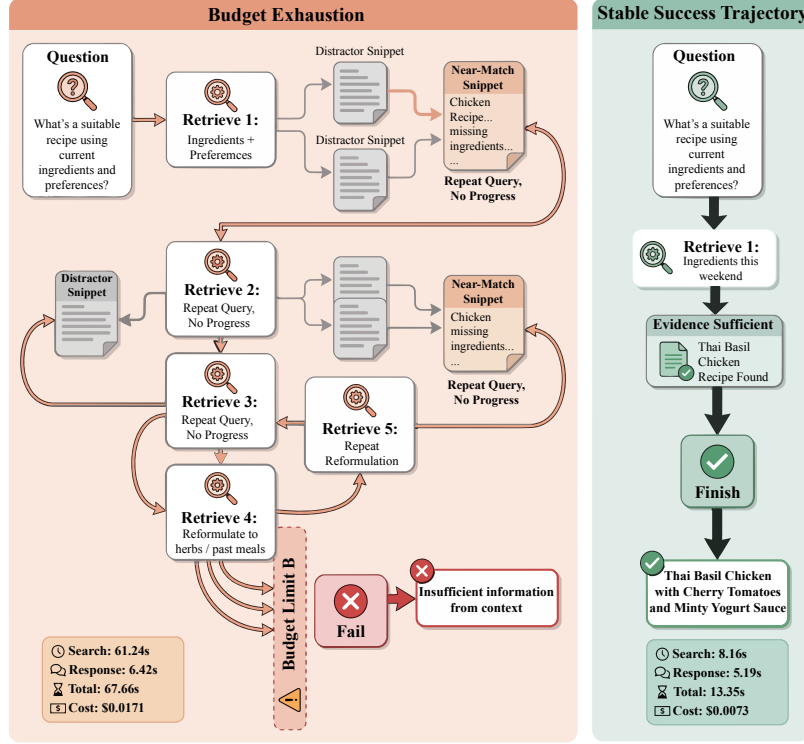


Figure 8: **Matched qualitative trajectories on the same LongMemEval item under LiCoMemory at $s = 400$.** The weaker model fails by exhausting the retrieval budget after repeated retrieve–reformulate steps, whereas the stronger model reaches the answer after one effective retrieval on the same item.

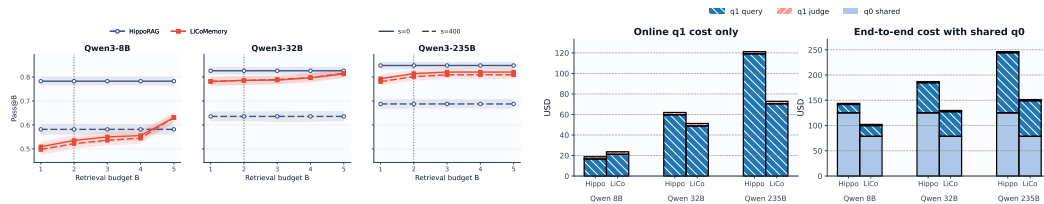


Figure 9: **Additional diagnostics on representative LongMemEval settings.** Left: budgeted-reliability curves $\text{Pass}@B$ under $B \in \{1, \dots, 5\}$. HippoRAG remains essentially flat because it is single-pass, whereas LiCoMemory recovers as the budget is relaxed, especially for smaller models. Right: q_0/q_1 cost accounting. This panel separates shared preprocessing cost from online query-time cost and clarifies that the paper’s retrieval-budget analysis targets the online agent-facing burden rather than denying offline cost differences across systems.

418 A.5 Detailed Memory Taxonomy

419 A.6 Flat and Planar Parity Conditions

420 The protocol is defined for flat, planar, and hierarchical memory interfaces. To avoid conflating
421 memory scale with implementation-specific access differences, any flat or planar run reported under
422 this protocol must satisfy the same parity constraints as the hierarchical runs. Specifically, for each
423 query q and scale s , the accessible history $\mathcal{H}^{(s)}$ must contain the same task-relevant evidence and
424 the same injected irrelevant sessions; the retrieval-call budget must be $B_0 = 2$ for primary results;
425 the scale levels must be s_0-s_4 with $N_{\text{irr}} \in \{0, 100, 200, 300, 400\}$; and each retrieval action must
426 return the same number of items. For dense or single-pass systems this is the returned top- k list; for
427 graph or multi-stage systems it is the final set returned after internal traversal or reranking. Systems

Table 10: **Detailed memory taxonomy used by the unified evaluation protocol.** Flat, planar, and hierarchical systems differ in the structure exposed to the agent at retrieval time, but are evaluated under the same scale ladder and retrieval-call budget; when reported, we also match the number of returned items per retrieval step across systems.

ID	Method	Subtype	Store unit (granularity)	Retrieve	Update?
F1	LightMem	Summary	summarized chunk	constrained embedding-based semantic retrieval	soft/offline time-augmented merge
F2	MemOS	Chunk	chunk	hybrid retrieval	merge
P1	A-Mem	Graph	memory note	top- k graph-aware retrieval	overwrite / memory-note update
P2	LinearRAG	Relation-free graph	passages linked through entity and semantic-link nodes in a Tri-Graph	two-stage graph retrieval: entity activation via local semantic bridging, followed by passage selection through global importance aggregation	offline graph construction / graph refresh
H1	HippoRAG	Hierarchical graph	multi-granularity graph memory (passage / fact / entity)	single-pass hybrid retrieval with graph propagation	graph augmentation / refresh
H2	LiCoMemory	Cognigraph (hierarchical graph)	entity nodes + relation edges with chunk/session provenance	entity-led graph retrieval + temporal-semantic reranking	incremental node/edge merge

may differ in chunking, indexing, linking, traversal, reranking, and update policy, but not in the underlying evidence, irrelevant-session set, or per-query retrieval budget.

Under these constraints, flat systems such as LightMem and MemOS, planar systems such as A-Mem and LinearRAG, and hierarchical systems such as HippoRAG and LiCoMemory are evaluated as different memory interfaces over the same agent-facing memory-scale condition. Therefore, comparisons across F/P/H systems should be interpreted as interface comparisons, not as changes in task difficulty or memory-store content.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”,**
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers.**

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [TODO]

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The authors should answer **[Yes]** if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer **[No]**, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or **[No]**, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.